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The Emotion Engine

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Abstract

Language inevitably compresses and simplifies human emotion, causing subtle affective nuances to be lost in verbal expression. Although large language models (LLMs) can recognise and generate emotional language, most existing methods rely on discrete emotion labels or token-level control, which limits fine-grained and intuitive affective manipulation.

This project proposes an Emotion Engine that represents emotion as a continuous latent vector, rather than explicit linguistic categories. Given an input text, the system extracts its underlying emotional state as a low-dimensional emotion vector, which composes multiple overlapping affective directions. Users can directly adjust intensities of those vectors before regenerating text, that reflects the edited emotional state. Importantly, this operation is performed prior to language generation, bypassing the lossy compression inherent in verbalisation.

By leveraging recent advances in latent-space reasoning and internal activation steering in LLMs, the Emotion Engine enables controllable, pre-verbal emotion editing within a text-to-text pipeline. Human communication is always mediated by media. This project can affect all communication in the sense that it deals with feelings evoked by the media. Recent creative act with LLM prompt engineering may in fact be a role reversal, since creativity should serve before the verbalisation. By making it possible to edit the feelings that were really wanted to be conveyed, this project could bring about a more precise communication that society has never seen before.

Acknowledgments

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Chapter 1

Introduction

1.1 Words draw boundary lines across the ocean of “concepts”

When I was still in elementary school, my favourite biography in the library was Helen Keller’s. Stricken by illness as a child, she lost both sight and hearing and bore the “triple hardship” of being blind, deaf, and mute, yet devoted herself to the advancement of education and welfare for the disadvantaged.

But Helen Keller’s story is not merely a tale of “overcoming adversity.” What is etched most vividly in my memory is the moment when she first understood the word “water.” At first, Helen insisted that “cup” and “the water in it” were the same thing. However, the instant when Anne Sullivan signed “W-A-T-E-R” into her hand while letting the cold water from a pump run across Helen’s palm, her world changed. When she could separate the object that flows from the container, the continuum of chaos was reshaped into an ordered world with partitions.

This scene exemplifies the essence of language. We do not grasp the external world directly as “things-in-themselves.” The information flowing into our senses is, in truth, a continuum; language draws borders there and carves out units of meaning. Even the boundary between river and sea is not sharply inscribed by nature itself; it is defined for human convenience and by culture – “this far is river,” “from here is sea.” Words do not merely copy lines already inscribed upon the world; they are the very act of drawing those lines as needed. In other words, language is not just a label: it is the activity by which we impose boundaries within our cognition and shape the world we see. To learn words is not merely to expand vocabulary; it is to gain a way of partitioning the ocean of concepts and choosing how to look at the world. In that sense, we are, day by day, sketching new boundaries.

1.2 Verbalisation is a lossy compression

Verbalisation is an act of forcing the complex and continuous sensation into discretised vocabulary and grammar. When we perform **verbalisation**, there lurks a commonly overlooked risk; it is intrinsically **lossy**. The moment we verbalise, the fine nuances contained within raw feeling are abstracted into the fences of cognition that humanity has drawn at the level of words. Even if we choose the same term, the warmth that term carries depends on each person’s embodied history, and it is eventually averaged out. The relief or pride we felt when we opened our old school yearbook for the first time in a decade, once put into text, may be flattened into a bland “how nostalgic.” Moreover, lexicons and meanings for emotions differ greatly across languages and cultures[1]; terms like “anger” or “joy” do not necessarily point to the same nuances around the world. And yet humanity cannot let go of language. Even if it is a lossy compression, language remains the only standardised interface that connects the human heart to society.

1.3 Objectives

The goal of this project is not to deny language but to design a parallel circuit that supplements the components lost in the compression of verbalisation, i.e. a circuit for restoring what words leave behind. What if we could scoop up the human central cognition directly, without passing through the rendering pipeline of language, and translate it into other forms of expression? Could we transfer the lingering sensation that arises when reading a letter from your significant other into poetry or essays? Or could we improve novel translations by capturing emotions lost through differences in lexical definitions between languages? Here lies an uncharted gap in our understanding of human cognition.

This project aims to emulate a kind of human synaesthesia. In other words, we aim to map emotions across modalities. We will develop the foundational technology to extract, manipulate, and re-synthesise a **pre-verbal, intermediate representation of emotion**, and explore possible ways to steer the model output using those representations of emotion.

The final product of this project is a computational system capable of extracting, manipulating, and presenting the emotions contained in input text without having to go through the compression of verbalisation. The system focuses not on what the text superficially tells, but on the emotional state that existed immediately before the text was produced, and treats it as a continuous vector representation.

When a user enters any text, the system analyses the emotional nuances contained in that text and visualises them as a low-dimensional continuous vector. These emotion vectors are not discrete labels such as “joy” and “sadness,” but are expressed as multiple overlapping emotional directions, thus preserving subtle mixed emotions and fluctuations that cannot be captured by language.

Users can intuitively manipulate this emotion code space. For example, by strengthening only a specific emotional direction and weakening another direction for the simultaneous presence of relief and loneliness in a certain sentence, it is possible to sensitively edit the emotional state that one really wanted to convey. It is important to note that this operation is not performed at the linguistic level, such as vocabulary selection or stylistic adjustment, but rather on the emotional expression itself before the language is generated.

The edited emotion code is again transformed into a linguistic expression. The text generated here is not simply a paraphrased sentence, but is the result of a re-expression that retains or intentionally transforms the emotion contained in the original sentence.

In this process, the user does not need to put their emotions into explicit words. The verbal averaging and simplification that is usually considered inevitable between emotion recognition and generation is bypassed, and the raw emotion felt by the user can be transferred to another form of expression in a more controllable manner.

This end product is neither a system for classifying and diagnosing emotions nor a tool for automatically generating emotional sentences. Rather, it is an interface that temporarily holds the emotions that people had before they put them into words as manipulatable intermediate expressions, from which new linguistic expressions can be launched. The ultimate value of this project lies in the fact that it makes it possible to return to the boundaries drawn by language and choose words again.

Chapter 2

Background

2.1 Emotion Representation in LLMs

2.1.1 Discrete Emotion Labels and Their Limitations

Early work treated emotion as a small set of discrete categories. The most influential taxonomy is Ekman’s seven basic emotions: anger, contempt, disgust, enjoyment, fear, sadness, and surprise [2].

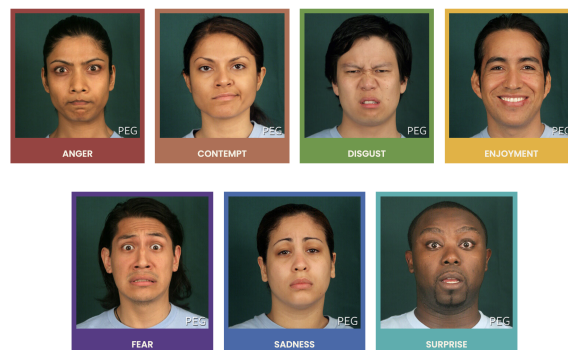


Figure 2.1: Paul Ekman’s seven basic emotions [3]

This assumes that emotions are biologically universal and psychologically irreducible. This framework enabled early supervised emotion classification systems by providing a clear labeling scheme and facilitated the construction of benchmark datasets. However, the categorical assumption introduces strong inductive bias; emotions are treated as mutually exclusive and qualitatively distinct.

Subsequent datasets attempted to increase expressive granularity. GoEmotions [4], for example, expanded the label space to 27 emotion categories and allowed multi-label annotation.

Positive		Negative		Ambiguous
admiration 🌟	joy 😄	anger 😡	grief 😞	confusion 😕
amusement 😂	love ❤️	annoyance 😠	nervousness 😰	curiosity 😟
approval 👍	optimism 🙌	disappointment 😞	remorse 😔	realization 💡
caring 🤗	pride 😊	disapproval 🗨️	sadness 😢	surprise 😲
desire 🍷	relief 😌	disgust 🤢		
excitement 🥳		embarrassment 😳		
gratitude 🙏		fear 😨		

Figure 2.2: GoEmotions taxonomy: 28 emotion categories including “neutral”. [5]

This enabled us to capture cases where multiple emotions occur in a single utterance. It is reported that emotions cluster by sentiment polarity and intensity, and that many categories exhibit high inter-label correlation. Nevertheless, despite its increased resolution, the dataset still relies on one-hot / sparse multi-hot encodings. Such encodings collapse emotional intensity and compositional structure into discrete indices, preventing the representation of graded transitions. For example, the difference between mild and intense anger cannot be represented, and the blend of emotions, such as bittersweet emotion, is also difficult.

Essentially, discrete labels force the learning problem into a classification regime that cannot capture continuity, geometry, or arithmetic structure. For example, two samples labeled joy may be emotionally distant, while joy and contentment may be emotionally close; yet this structure is invisible to the model itself. These limitations motivate a shift away from purely categorical representations toward continuous emotion spaces.

This problem of the discretised emotion labels is also raised by Wang and Zong (2021) [6]. They explicitly formalise this by arguing that one-hot emotion labels ignore latent relationships between emotion categories. They observe that emotion calsses are not orthogonal in human perception, but rather exhibit similarity in a gradation manner. More broadly, discrete labeling schemes prioritises the annotation convenience over the representational adequacy. While it simplify supervision and evaluation, they impose artificial bounadies that obscure the underlying structure of emotions. This mismatch becomes especially problematic when attempting cross-dataset transfer or cross-cultural comparison, where differences in label granularity cannot be reconciled within a purely categorical framework.

These shortcomings have motivated a growing body of work that seeks to represent emotion in continuous spaces, where distance, direction, and magnitude encode affective relationships. By having a representation methods that preserves intensity, similarity and compositionality, we can enable models to capture nuanced emotional variation beyond rigid class boundaries. This transition from discrete labels to continuous emotion spaces is the focus of the next subsection.

2.1.2 From Discrete Labels to Continuous Emotion Spaces

Continuous emotion representations seek to encode emotion as points in a low-dimensional vector space, where distance, direction, and magnitude correspond to psychological relationships. Early psychological models such as the pleasure–arousal–dominance (PAD) framework [7] formalised emotion along continuous axes, suggesting that emotions differ quantitatively rather than categorically.

1. **Pleasure** represents the hedonic quality of an experience, ranging from unpleasant to pleasant. States such as joy, contentment, and satisfaction occupy the positive end of the pleasure axis, whereas sadness, disgust, and despair lie toward the negative end.
2. **Arousal** captures the degree of physiological and cognitive activation associated with an emotional state, ranging from calm or lethargic to excited or agitated. High-arousal states include fear, anger, and excitement, while low-arousal states include relaxation, boredom, or melancholy.
3. **Dominance** reflects the extent to which an individual feels in control of, or overpowered by, a situation. High-dominance states are associated with feelings of agency, confidence, and control (e.g. pride, anger), whereas low-dominance states correspond to submission, helplessness, or vulnerability (e.g. fear, shame).

Together, these three dimensions define a continuous affective space in which canonical emotion categories correspond to regions rather than points. For example, anger and fear are both negatively valenced and highly aroused, but are separable along the dominance axis; sadness and boredom share low arousal and negative valence, but differ in dominance and intensity. This geometric interpretation naturally accounts for graded emotional intensity, mixed emotions, and smooth transitions between affective states – phenomena that are difficult to represent with discrete labels.

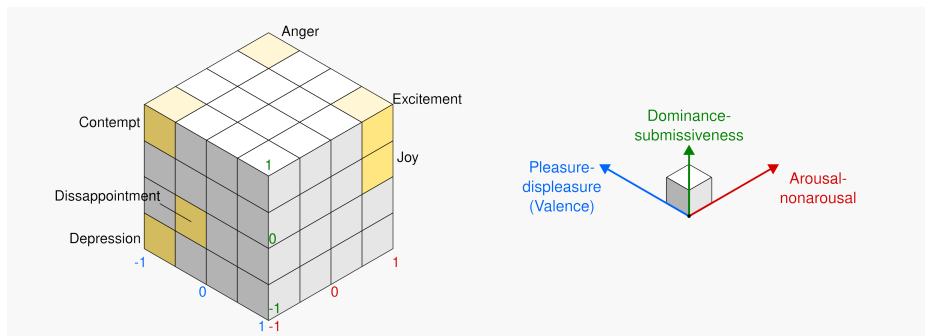


Figure 2.3: Representation of PAD model done with cubes [8]

Additionally, in the field of affective computing, an explicit attempt to bridge the gap between discrete labels and continuous emotion representations is presented

by Buechel et al. [9]. Rather than committing to a single emotion taxonomy, they propose learning *label-agnostic emotion embeddings* that serve as a shared latent representation across heterogeneous annotation schemes, including polarity, basic emotion categories, and affective dimensions such as valence and arousal.

Crucially, emotion labels in this framework are not treated as targets to be predicted directly, but as projections from a common embedding space. Models are trained to map text into this latent affective space, from which multiple label formats can be recovered via lightweight prediction heads. This design enables the learned emotion information to be transferred to another without retraining the full model. Empirically, Buechel et al. show that such embeddings preserve prediction quality while supporting cross-dataset generalisation.

Complementary to this, Wang and Zong (2021) [6] focus on learning continuous representations for *emotion categories themselves*. Instead of representing emotion classes as one-hot vectors, they derive distributed emotion embeddings by averaging soft classifier outputs over all instances of a given emotion category. These representations form an emotion space in which distances reflect empirical similarity between emotions as expressed in text. Their results demonstrate that emotion categories cluster by sentiment polarity and intensity, which captures the emotions more faithfully than the conventional semantic word embedding spaces.

Both of those approaches we looked at enable emotion to be represented as a continuous, low-dimensional structure where each of them corresponds to a region or a direction in a metric space. This perspective aligns closely with psychological theories such as PAD, while remaining grounded in data-driven learning from natural language.

Importantly, representing emotion in continuous spaces shifts the learning problem away from rigid classification toward geometric representation learning. This not only resolves many of the limitations inherent in discrete labels, but also provides a foundation for analysing how emotional information is encoded in vector representations more generally. In particular, it raises the question of whether emotion occupies identifiable subspaces or linear directions within learned embeddings, a topic explored in the following subsection.

2.1.3 Linear Structure of Emotion in Word Embedding Spaces

An important question here is whether the continuous space for emotion is actually reflected in learned language representations. Interestingly, prior to the emergence of LLMs, static word embeddings already exhibited evidence of latent affective organisation, despite being trained without explicit emotional supervision.

Wu and Jiang [10] demonstrate that emotional information in word embeddings such as GloVe [11] is not uniformly distributed across dimensions, but instead concentrated along a small number of linear directions. They train a linear autoregressive model to predict continuous emotional valence on annotated narrative data, and show that only a limited subset of embedding dimensions contributes to emotional

valence. Interpreting the learned weights reveals that emotional polarity can be captured by a low-dimensional subspace embedded within the full semantic space.

It is also reported that projecting word embeddings into this learned emotion subspace produces a substantially clearer separation between positive and negative emotion. This indicates that emotion is linearly recoverable even when embeddings are trained without affective objectives. Namely that, emotional meaning is encoded **geometrically**, rather than a byproduct of lexical associations.

A particularly significant finding is that this projected emotion space preserves *linear compositional structure*. Vector arithmetic over words that contain emotion yields composite affective states, which is also consistent with psychological theories of emotion composition such as Plutchik’s wheel of emotions [12].

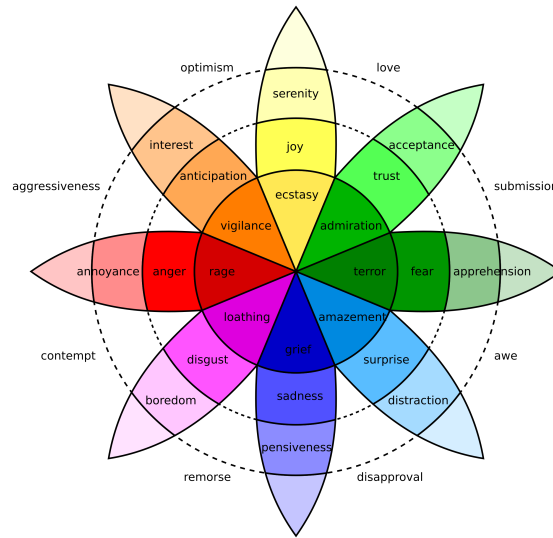


Figure 2.4: Plutchik’s wheel of emotions: basic emotions are arranged radially, with angular proximity indicating similarity and radial intensity indicating emotional strength.

For example, the vector sum of joy and trust aligns closely with love, while remaining distant from affectively opposing states such as remorse. This behaviour cannot be captured by categorical representations, but emerges naturally in a continuous, vector-valued framework.

These results indicate that emotion in word embedding spaces is not only continuous but also approximately linear, admitting meaningful projections, directions, and arithmetic operations. As a result, affective similarity, intensity, and composition can be expressed through simple linear operations rather than discrete class boundaries.

Taken together, these findings provide early evidence that emotion is encoded as structured geometry within learned language representations. This establishes a representational bridge between psychological models of emotion and mathematical embedding paradigms. This also motivates further research into how such linear emotional structure scales and evolves in more expressive representation learning systems.

2.2 Manipulation of Latent Representations

To harness these emotion vectors, researchers have explored techniques for directly manipulating internal representations of models.

2.2.1 Neuron-Level Emotion Control

One line of work focuses on exploiting localised affective representations at the neuron level. For instance, Radford et al. (2017) [13] noted a single “sentiment neuron” in a byte-level recurrent language model. It has shown that the activation strongly correlates with review polarity and the manipulation of that sentiment neuron directly controls the generated sentiment. Later work shows that modern LLMs contain clusters of neurons selectively responsive to specific emotions.

Ablation and masking experiments were done by Lee et al. (2025) [14], and it demonstrates that removing these neurons can selectively impair recognition of particular emotions. This confirms that they play a functional role rather than being incidental correlates. However, there is still a possibility that the model is redundant. For some emotions, performance remains stable after ablation, suggesting overlapping representations and compensatory mechanisms.

Additionally, low-resource fine-tuning methods such as LoRA (Low-Rank Adaptation) [15] and prefix-tuning operate at a higher level of abstraction but still manipulate latent activations. By injecting small learned vectors into the model we can induce a consistent emotional tone without full retraining. This shows that affective style is encoded in directions that are easy to steer with low-rank updates.

2.2.2 Emergent Emotional Subspaces in Large Language Models

Large language models substantially extend and generalise the affective structure previously observed in static word embeddings (2.1.3). Even though models are trained on massive corpora without any emotional information supervision, LLMs acquire rich contextual representations. These models appear to internalise emotion as a continuous and distributed property of their hidden-state geometry.

Recent work provides strong empirical evidence for this claim. Reichman et al. (2025) [16] show that emotional information in large decoder-only LLMs is organised within a low-dimensional **emotional manifold** embedded in the activation.

They analysed hidden states across layers, demonstrated how affective content aligns with a small number of stable latent directions. Importantly, these directions appeared without any explicit emotion supervision, indicating that affective structure is a natural consequence of large-scale language modelling rather than a task-specific artefact. Moreover, the identified emotional subspace is remarkably consistent across domains and languages, suggesting that LLMs internalise a broadly universal affective geometry.

Crucially, this geometry is not merely descriptive. Reichman et al. [16] show that small interventions along these latent directions can alter the model’s emotional interpretation of an input, while preserving its semantic content. This establishes emotional subspaces as functionally meaningful components of the model’s representation space, rather than passive correlates of surface-level sentiment cues. This establishes emotional subspaces to be considered as a functional and meaningful components of the model’s representation space.

Complementary evidence comes from neuron-level analyses too. Lee et al. (2025) [14] identify groups of neurons whose activations selectively track specific basic emotions in large instruction-tuned models. Emotional information is broadly distributed, but certain neurons constantly exhibit disproportionately strong responses to particular affective states. Further ablation experiments revealed that suppressing these neurons can significantly degrade emotion recognition performance for some emotions. This provides a causal evidence that emotional information is actively processed rather than epiphenomenal.

Taken together, these findings suggest that emotion in LLMs is best characterised as an emergent, low-dimensional subspace embedded within high-dimensional contextual representations. Emotional meaning is not localised to single units, nor reducible to discrete labels; instead, it occupies a structured region of stable latent space. This emergent affective geometry provides the representational foundation that enables direct control of emotional behaviour through linear interventions, which we discuss next.

2.2.3 Steering Latent States with Linear and Continuous Control Vectors

If emotional meaning in LLMs is encoded as directions or subspaces within latent representations, a natural consequence is that affective behaviour should be directly controllable by intervening along those directions. This observation has motivated a line of work on **latent steering**, in which linear and continuous control vectors are applied directly to internal activations during inference.

An early and influential approach is Plug-and-Play Language Models (PPLM) [17], combines a frozen language model with an external attribute model (e.g. a sentiment classifier) and performs gradient ascent on the hidden states to increase the likelihood of a desired attribute. Effectively, this method introduces a small **sentiment vector** into the latent representation, biasing generation toward positive or negative emotion without modifying the model’s weights. Although originally framed as a controllable generation technique, PPLM implicitly relies on the assumption that affective attributes correspond to approximately linear directions in latent space.

More recent work makes this assumption explicit and removes the need for iterative optimisation. Anthropic’s **Contrastive Activation Addition** (CAA) [18] directly intervenes in a model’s internal activations using precomputed **steering vectors**. These vectors are obtained by collecting large sets of positive and negative examples

of a target behaviour or attribute and averaging the differences in their intermediate representations. The resulting vector captures a direction in latent space that causally corresponds to the desired attribute. During inference, adding this vector to the residual stream steers model behaviour in a predictable manner, while leaving the model parameters unchanged.

A key advantage of CAA is its simplicity and interpretability. The intervention is linear, and the strength of the effect can be modulated continuously by scaling the steering vector with a scalar coefficient. Moreover, multiple steering vectors can be combined additively, enabling compositional control over several attributes simultaneously. In contrast to prompt engineering or fine-tuning, such interventions operate directly on the model’s internal state and often yield more precise and robust control, particularly for abstract properties such as sentiment or emotional tone.

Crucially, steering latent states does not only affect surface-level lexical choices. Injecting affective control vectors at intermediate layers can alter the model’s internal interpretation of emotional context, changing how emotional cues are integrated and propagated through the network. This supports the view that emotion in LLMs functions as a global latent variable, rather than as a property tied to individual tokens or outputs.

As conclusion, latent steering methods such as PPLM and CAA provide strong evidence that emotional subspaces in LLMs are not only emergent and structured, but also causally actionable. Manipulating affective behaviour through simple linear interventions reinforces the interpretation of emotion as a continuous geometric representation space. It further bridges descriptive analyses of emotional geometry with concrete mechanisms for controlling model behaviour.

2.2.4 Causal Evidence for Linear Sentiment and Emotion Directions

The existence of affective directions in representation space can be shown by linear probes or dimension reduction, however, it is still doubtful if they actually play a causal role in model behaviour. Recent work provides strong evidence that emotion directions in LLMs are not only linearly recoverable, but also functionally necessary and sufficient for affective reasoning.

Hollinsworth et al. (2024) [19] show that sentiment in LLMs is encoded along a single dominant linear direction in contextual embeddings. They perform directional ablations and interventions, demonstrating that removing / suppressing this sentiment direction causes systematic degradation in downstream emotion-sensitive tasks such as emotion classification. Conversely, injecting the same direction into neutral contexts induces predictable shifts in model outputs. This bidirectional manipulation provides an evidence that the emotion direction is actively used by the model rather than being a passive correlate.

Importantly, affective competence in LLMs does not depend on explicit fine-tuning for emotion-related tasks. Broekens et al. (2023) [20] demonstrate that off-the-shelf

LLMs such as ChatGPT can map text to precise valence–arousal–dominance (VAD) values and detect fine-grained emotional cues using prompting alone. This suggests that continuous affective representations are already embedded in the latent space as part of general language understanding. This is a strong evidence that interprets emotion as a pre-verbal latent variable, rather than a task-specific output format.

At larger scales, evidence further indicates that emotional representations are not flat but hierarchically structured. Zhao et al. (2024) [21] observe that very large models (more than 405 billion parameters) develop tree-like emotion representations, where basic affective states branch into more specific emotions. Models featuring a more consistent hierarchical affective geometry demonstrate enhanced emotion recognition capabilities and increased persuasive effectiveness in negotiation scenarios. This directly links the internal structure of emotional expression to functional behaviour.

These findings converge on a causal interpretation of affective geometry in LLMs. Removing emotion directions impairs affective reasoning, while injecting them predictably alters behaviour. Moreover, emotional information is not confined to isolated tokens or neurons, but is accumulated and summarised across layers into global latent variables that shape general affective capabilities of models.

In conclusion, there is a principled foundation for vector-based manipulation of emotion, which enables us to express emotional nuance through numeric operations on embeddings rather than token-level substitutions. This causal grounding is essential for our project that seeks to control and align emotional behaviour in large language models.

2.3 Reasoning Beyond Token Space

2.3.1 Limitations of Token-Based Reasoning

Conventional LLMs reason in token space. Because inputs are projected onto the discrete boundaries humanity drew as “vocabulary,” fine affective nuance and embodied subtleties are irretrievably lost during that projection, and what remains in one output language may differ from what remains in another. To adjust tone, one often uses prompt engineering or instruction tuning (“Please answer politely,” etc.), but adherence is uncertain and fine-grained control is difficult. Other countermeasures include supervised fine-tuning to internalise a style, or label-conditioned generation (e.g. adding <joy> <sad>), but fine-tuning is costly and switching between styles is inflexible; label conditioning also cannot capture complex blends of nuance.

Beyond affective control, the same limitation applies to reasoning itself. Autoregressive language models do not take token functionality into account in its generation process. Even if the token carries significant reasoning content, it will be allocated approximately same computational budget as grammatical tokens like *a* or *is*. As a result, only a small subset of internal states carries the actual inferential work, and

most of the tokens in explicit reasoning traces do not contribute to the underlying computation.

Recent works show this disconnectivity between the surface output tokens and internal reasoning process. Pfau et al. (2024) [22] demonstrate that transformers can perform non-trivial hidden computation even when output tokens are semantically uninformative. For example, inserting filler or pause tokens (carry no explicit meaning) somehow measurably improve performance on reasoning tasks. This suggests that the essential computation is carried by latent activations rather than by the emitted tokens themselves. Token generation only a communicative language constraint around an internal process.

The inefficiency of token-based reasoning becomes more apparent when reasoning is explicitly verbalised. Chain-of-thought prompting improves performance in many tasks, however **it requires the model to externalise intermediate states into natural language, even when those states are not naturally linguistic**. As argued by Hao et al. (2025) [23], language is optimised for communication rather than for computation. Forcing reasoning to unfold within the constraints of a discrete vocabulary introduces unnecessary overhead and distortion. Many reasoning tokens exist not because they are required for inference, but because they are required for human readability.

From this perspective, explicit verbalisation constitutes a form of lossy compression for an affective reasoning as discussed in the previous section. The continuous and high-dimensional internal states during inference must be projected onto a discrete and culturally contingent token space, which flattens distinctions that are visible for computation but words-level invisible. This compression is problematic for affective computing, because emotion structure is inherently graded, which is resistant to precise lexicalisation.

Taken together, these observations suggest that token space not a substrate of reasoning, but rather an interface. While it provides a convenient medium for communication and supervision, it is suboptimal for internal computation. This motivates a growing line of work that seeks to decouple reasoning from explicit token generation. Computation can proceed without the constraints imposed by language.

2.3.2 Reasoning in Continuous Latent Space

Motivated by the observation that reasoning is primarily carried by latent activations rather than surface tokens, a growing body of work has explored **implicit** forms of chain-of-thought. The central idea is to retain the intermediate reasoning within the model’s hidden state, so that we can benefit from the multi-step reasoning while avoiding the distortions introduced by forcing intermediate states into natural language.

Early implicit chain-of-thought approaches aim to internalise reasoning traces by gradually removing explicit reasoning tokens during training, encouraging the model to perform the same computation silently in its latent space. More recent methods

go further by explicitly allocating computation to continuous latent representations that replace textual reasoning steps altogether. In these models, reasoning unfolds as a sequence of hidden-state transitions, while the output layer is used only to produce the final answer.

A representative work is **CODI** (Compressing Chain-of-Thought into Continuous Space via Self-Distillation) [24]. This approach compresses explicit chain-of-thought trajectories into a compact sequence of continuous latent states, and by distilling a teacher model that produces explicit reasoning traces, CODI trains a student model to reproduce the same reasoning outcome using a shorter sequence of latent vectors. This demonstrates that the information carried by textual reasoning can be encoded more efficiently in continuous space, without requiring token-level supervision at inference time.

A more radical approach is proposed by **Coconut** (Chain of Continuous Thought) [23]. Rather than compressing explicit reasoning traces, Coconut modifies the inference procedure itself by feeding the model’s last hidden state back as the next input embedding. In this way, reasoning unfolds entirely within continuous latent space and there is no need of intermediate textual tokens at all. However, because Coconut relies primarily on answer-level supervision, the resulting latent trajectories can be unstable when extended to deeper reasoning chains, which we discuss in the next subsection.

2.3.3 Implicit and Continuous Chain-of-Thought

However, there is one challenge that purely implicit reasoning introduces. When multiple latent reasoning steps are chained together without sufficient structural constraints, the latent representations can collapse into homogeneous states. This phenomenon is often referred to as **latent collapse**, which undermines the model’s ability to maintain distinct intermediate representations.

SIM-CoT (Supervised Implicit Chain-of-Thought) [25] addresses this limitation by introducing step-level supervision during training. Instead of supervising only the final answer or the overall reasoning trajectory, SIM-CoT attaches an auxiliary decoder that forces each latent reasoning step to predict a corresponding explicit reasoning phrase. This alignment ensures that each latent vector encodes distinct, semantically meaningful content, preventing homogenisation and improving stability. Crucially, the auxiliary decoder is used only during training and is removed at inference, allowing the model to retain the efficiency of silent latent reasoning while benefiting from structured supervision. Empirically, SIM-CoT substantially improves both performance and robustness over earlier latent reasoning models, including Coconut-style architectures.

It attaches an auxiliary decoder that forces each latent reasoning step to predict a corresponding explicit reasoning phrase, which ensures that each latent vector to encoded semantically meaningful content. This substantially improves both performance and robustness over earlier latent reasoning models, including Coconut-style

architectures. Importantly, the auxiliary decoder is used only during training and is removed at inference, allowing the model to retain the efficiency of silent latent reasoning while benefiting from structured supervision.

These approaches demonstrate that chain-of-thought need not be realised as a sequence of discrete tokens. Reasoning can be instead implemented as a controlled trajectory through continuous latent space, with explicit language serving primarily as a training scaffold or interpretability aid. Implicit and continuous chain-of-thought models provide a natural foundation for studying how emotions may be represented and manipulated within latent reasoning processes.

2.3.4 Implications for Affective Representation

The developments reviewed above suggest a converging picture of representation and computation in LLMs. As discussed in earlier sections, emotions are not confined to discrete labels or lexical choices but it rather emerges as a continuous structure within latent space. At the same time, recent advances in implicit chain-of-thought demonstrate that reasoning itself can be implemented as a trajectory through that same latent space, without requiring intermediate symbolic tokens.

These findings summarises to the fact that emotion need not be treated as separate components at the level of surface language. Instead, it appears to be grounded in shared latent representations that evolve over the course of inference. From this perspective, emotional information may constitute an integral part of the internal reasoning state itself, shaping how intermediate representations are formed and propagated.

Importantly, this possibility remains largely unexplored. Although latent reasoning frameworks have demonstrated improved expressivity for logical and mathematical tasks, their implications for affective representation have not been systematically examined. Whether emotional dimensions align or interact with latent reasoning trajectories is an open question. Addressing this gap requires moving beyond token-level analysis and considering emotion as a first-class component of latent computation rather than as a stylistic attribute.

These considerations motivate the investigation in the following chapters. Rather than treating emotion as an external label or an output constraint, we explore the hypothesis that affective states can be represented within continuous latent spaces that underlie reasoning itself. By doing so, we aim to better understand how emotional meaning may be re-expressed when language is no longer the primary medium of inference.

Chapter 3

Dev: Affective Steering Pipeline

The purpose of this chapter is to demonstrate that we can construct an emotional direction within the residual space of a language model, and use it to actually steer the output. We do not delve deeply into the semantic interpretation of the emotion representations themselves yet. Rather, our empirical claim is that, for a given target emotion e , applying a CAA direction derived from contrastive residual differences can shift the affective framing of the output towards the target emotion while largely preserving semantic content.

The central conclusion of this chapter is that by decomposing emotion-specific CAA vectors into a shared emotionalisation component and an emotion-specific residual component, we can bias the output toward a target emotion. Namely that,

$$\Delta_e(\alpha_G, \alpha_R) = \alpha_G g + \alpha_R \hat{r}_e$$

where g is the shared emotionalisation vector that shifts from neutral to emotional expression, and $\hat{r}_e$ is the residual vector specific to the target emotion. In this chapter, we treat this decomposition as a practical and operational representation for steering, rather than making strong claims about the underlying semantics of the emotion representations themselves.

Experimental Roadmap

This chapter is organised as follows.

1. **Dataset construction:** From a single base utterance, we generate emotional rewrites corresponding to multiple target emotions and intensity levels, and further construct neutral paraphrases. This allows us to create a controlled corpus that enables the comparison of emotional variations while preserving the original meaning.

2. **Residual-stream extraction:** We combine an emotion rewrite and the corresponding neutral paraphrase into a prompt of the same format and extract residual-stream activations from selected layers of Llama-3.1-8B-Instruct. For each source utterance, we compute

$$\mathbf{H}_{n,e,i,l}^{\text{emo}} - \mathbf{H}_{n,l}^{\text{neu}}$$

to obtain an affective shift that cancels out the semantic content as much as possible.

3. **Layer selection and diagnostics:** We compare diagnostic metrics such as emotion-category probe accuracy and centroid compactness, to select the primary analysis layer. The goal here is not to determine the most interpretable layer, but rather to determine the layer that provides the most stable affective direction for steering.
4. **CAA vector construction:** We construct contrastive activation addition (CAA) vectors by averaging the residual vectors over sources and intensities, and then normalising to obtain unit vectors for each emotion.
5. **Operational decomposition for steering:** Since cross-emotion CAA directions exhibit very high cosine similarity, this suggests that each emotion direction contains a large shared emotionalisation component. Therefore, we define

$$\mathbf{g} = \frac{\frac{1}{|E|} \sum_{e \in E} \mathbf{c}_e}{\left\| \frac{1}{|E|} \sum_{e \in E} \mathbf{c}_e \right\|_2}$$

and remove this shared component from each emotion direction.

$$\hat{\mathbf{r}}_e = \mathbf{c}_e - (\mathbf{c}_e^\top \mathbf{g}) \mathbf{g}$$

In this chapter, we operationally use this $\hat{\mathbf{r}}_e$ as the **steering residual** responsible for the target emotion identity. However, the internal affective structure of $\hat{\mathbf{r}}_e$ will be discussed in the next chapter.

6. **Steering experiments:** We add the steering vector $\alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e$ to the residual stream and evaluate whether the generated output shifts toward the target emotion.

3.1 Dataset Construction

In order to investigate whether emotion is encoded as structured geometry in residual space, we constructed controlled rewrite datasets from shared source utterances. The primary dataset is an emotion-category rewrite set derived from two dialogue corpora: DailyDialog [26] and EmpatheticDialogues [27]. We selected dialogue data because conversational utterances provide compact emotional signals grounded in

context in everyday interpersonal situations, while keeping propositional content sufficiently constrained for controlled rewriting. For each source utterance, we used GPT-4.1-mini to generate rewrites for Plutchik’s eight basic emotions (joy, trust, fear, surprise, sadness, disgust, anger and anticipation) at three matched intensity levels (low, medium and high). This yields $8 \times 3 = 24$ affective variants per source text. A central constraint throughout the construction process was to preserve meaning: the underlying situation and factual content were held constant while only the affective framing was systematically varied.

Emotion	Utterance
Base	I waited for hours and nobody came.
Joy	Nobody came for hours, but I ended up enjoying the quiet time on my own.
Trust	Nobody came for hours, but I am sure they had a valid reason.
Fear	Nobody came for hours, and I started worrying that something might have gone wrong.
Surprise	Nobody came for hours, and honestly I did not expect that at all.
Sadness	For hours, nobody came. I guess that says enough.
Disgust	Nobody came for hours; that felt deeply inconsiderate.
Anger	Nobody came for hours, and I am really frustrated about wasting all that time.
Anticipation	Nobody came for hours, but I kept thinking they might still arrive any minute.

Table 3.1: Illustrative rewrites for all eight Plutchik emotion categories at matched medium intensity.

As a control condition, we constructed an additional neutral paraphrase baseline dataset. In this dataset, each source utterance has been rewritten into an affectively neutral but semantically equivalent form. This baseline provides a reference representation of content without targeted emotional colouring, enabling contrastive analysis between emotion-conditioned and emotion-suppressed activations.

Field	Example
Base utterance	I waited for hours and nobody came.
Neutral paraphrase	I waited for several hours, and no one arrived.
Purpose	Preserve events and meaning while removing affective phrasing.

Table 3.2: Illustrative example from the neutral-paraphrase control dataset.

Together, these datasets enable us to distinguish emotional signals from lexical-semantic content. This provides an empirical basis for the decodability, geometry and steering analyses presented in subsequent sections.

This design makes it possible to ask whether affective variation forms a consistent direction in activation space, rather than merely reflecting surface-level lexical differences between emotional words.

3.2 Residual-Stream Extraction

We extract hidden-state activations from an instruction-tuned language model (Llama-3.1-8B [28]) with no gradient updates. Our aim is to determine where affective information is represented in the model’s internal computations rather than in output tokens alone. The residual stream is therefore used as the primary analysis target, because it is the shared channel through which each transformer block accumulates and propagates semantic and affective information across layers.

To ensure that the inputs are comparable across the different conditions, each sample is formatted with a shared instruction prefix that is constructed from the same base utterance. Only the continuation text is changed, either through an emotion rewrite or a neutral paraphrase. Specifically, we use a user instruction in the form of a chat template that asks for a rewrite aligned with the speaker’s feelings, and then append the rewritten sentence as the assistant continuation. This matching protocol controls for variance at the prompt level and enables direct comparison of emotion-conditioned and neutral-conditioned activations.

Template Prompt

Please rewrite this so it aligns with my feelings more.
 Start straight from the rewritten text without any preamble, and only provide **one** rewritten version.
 Base: {base_text}

Table 3.3: Prompt format used for activation extraction.

Residual sampling involves forwarding batched inputs with hidden-state output enabled and collecting vectors at selected hook layers. For each target layer l , the last token’s hidden state is taken from the layer’s output, i.e.

$$\mathbf{h}_l = \text{hidden_states}[l + 1][:, -1, :]$$

Since tokenisation uses left padding, the index -1 consistently corresponds to the final real token in each sequence in the batch. Final-token representation is the state after the model has processed the entire rewritten utterance, making it a compact summary of the affective framing expressed by the continuation. Activations are sampled from layers $\{8, 10, 13, 16, 19, 22\}$, producing a multi-layer representation for every input.

The emotion-intensity extraction pipeline produces a 5D tensor:

$$\mathbf{H}^{\text{emo}} \in \mathbb{R}^{N \times E \times I \times L \times D},$$

where N is the number of source texts, $E = 8$ emotions, $I = 3$ intensity levels, L is the number of sampled layers, and D is the hidden dimension. The neutral baseline pipeline produces:

$$\mathbf{H}^{\text{neu}} \in \mathbb{R}^{N \times L \times D}.$$

Source IDs are sorted to align both tensors by index, allowing direct subtraction

$$\Delta \mathbf{H}_{n,e,i,l,:} = \mathbf{H}_{n,e,i,l,:}^{\text{emo}} - \mathbf{H}_{n,l,:}^{\text{neu}}$$

which isolates affective shifts while controlling for base semantic content.

3.3 Layer Selection and Diagnostics

Not all transformer layers provide equally useful representations for affective analysis. Emotion-related structure may be present at multiple depths, but different layers can privilege different properties of the representation. For example, one layer may provide strong emotion-category separability, while another may better preserve a compact low-dimensional geometry. Since the later analyses require both linear decodability and geometrically stable affective directions, we compared candidate layers using several complementary diagnostics.

Based on these diagnostics, layer 13 was selected as the primary layer for the main analyses. It is not globally best on every metric, but it provides the most convincing overall trade-off between category separability, low-dimensional compactness, and sensitivity to graded affective variation.

Layer	Emotion probe BAcc (4-D subspace)	Centroid EVR ₄	Mean intensity-direction correlation
10	0.6751	0.8876	0.0998
13 *	0.7033	0.8920	0.1336
16	0.6704	0.8958	0.1297

Table 3.4: Layer-wise diagnostic summary for candidate layers. Emotion-category probe balanced accuracy (4-D subspace), centroid compactness (EVR₄), and mean local intensity-direction correlation. Bold indicates the best value per column; asterisk marks the selected reporting layer.

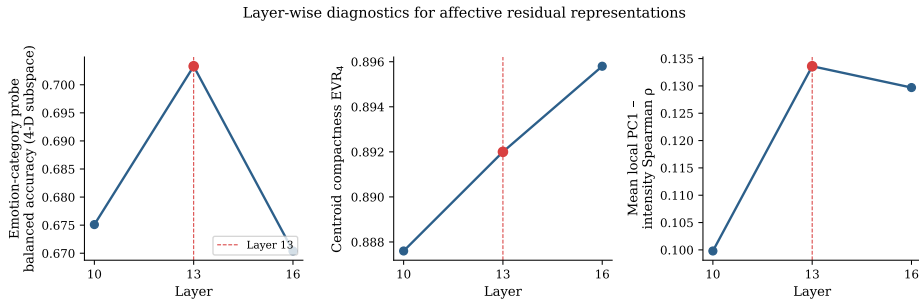


Figure 3.1: A visualisation of the layer diagnostics across candidate layers.

First, emotion-category probe performance in the 4D affective subspace is highest at layer 13. The balanced accuracy reaches 0.7033 at layer 13, compared with 0.6751

at layer 10 and 0.6704 at layer 16. On this criterion, layer 13 preserves the most linearly decodable information about the target emotion categories.

Second, this increase in decodability does not come at the cost of geometric compactness. The centroid-level four-dimensional explained variance remains high at layer 13, with $\text{EVR}_4 = 0.8920$. This is slightly below layer 16 ($\text{EVR}_4 = 0.8958$) but above layer 10 ($\text{EVR}_4 = 0.8876$). In other words, layer 13 is not the strongest layer on this metric alone, but it remains close to the best-performing depth while retaining stronger emotion-category separability.

Third, layer 13 shows the strongest sensitivity to graded affective variation among the candidate layers. The mean intensity-direction correlation is 0.1336 at layer 13, compared with 0.0998 at layer 10 and 0.1297 at layer 16. Although the absolute correlation is modest, the relative increase suggests that layer 13 is the most sensitive of the three candidate layers to systematic changes in affective intensity.

To avoid overfitting the analysis to a single depth, we also examined neighbouring layers as robustness checks. Layers 10 and 16 show similar qualitative trends and preserve the main geometric findings, suggesting that affective structure is not confined to a single transformer block. We therefore do not treat layer 13 as uniquely optimal in a global sense. Rather, we use it as the main reporting layer because it offers the best trade-off across the three diagnostics, while layers 10 and 16 serve as nearby robustness checks.

3.4 CAA Vector Construction

In order to convert contrastive residual differences into controllable steering signals, we derive Contrastive Activation Addition (CAA) directions from emotional and neutral conditioned activations. For each source utterance n , emotion e , intensity level i , and layer l , we first define

$$\mathbf{d}_{n,e,i,l} = \mathbf{h}_{n,e,i,l}^{\text{emo}} - \mathbf{h}_{n,l}^{\text{neu}}.$$

We then average over sources and normalise to unit length:

$$\bar{\mathbf{d}}_{e,i,l} = \frac{1}{N} \sum_{n=1}^N \mathbf{d}_{n,e,i,l}, \quad \mathbf{c}_{e,i,l} = \frac{\bar{\mathbf{d}}_{e,i,l}}{\|\bar{\mathbf{d}}_{e,i,l}\|_2}.$$

The vectors $\mathbf{c}_{e,i,l}$ are intensity-specific CAA directions.

As the steering stage primarily targets the emotional category, treating intensity as a secondary modulation, we also build pooled directions by averaging across intensities and re-normalising:

$$\mathbf{c}_{e,l}^{\text{pool}} = \frac{\frac{1}{I} \sum_{i=1}^I \bar{\mathbf{d}}_{e,i,l}}{\left\| \frac{1}{I} \sum_{i=1}^I \bar{\mathbf{d}}_{e,i,l} \right\|_2}.$$

In the main steering analyses, these pooled unit vectors are used as the primary candidate steering vectors.

Empirically, this construction remains stable at the selected reporting depth (layer 13). Alignment between local contrastive shifts and global CAA directions is high for all emotions at layer 13, with mean cosine values ranging from 0.8536 to 0.8570 and standard deviations of around 0.054. This is slightly stronger than layer 10 and much stronger than deeper candidate layers, such as layer 16 and above. This matches the result of the layer selection that layer 13 provides the best overall trade-off for controllable affective structure.

The pooled directions are also highly consistent across intensity levels. At layer 13, the cosine similarity between low, medium, and high CAA vectors remains very high for all emotions (approximately 0.983–0.999 across all intensity pairs). This suggests that intensity changes mainly in terms of magnitude around a shared category direction, rather than substantially altering the direction itself.

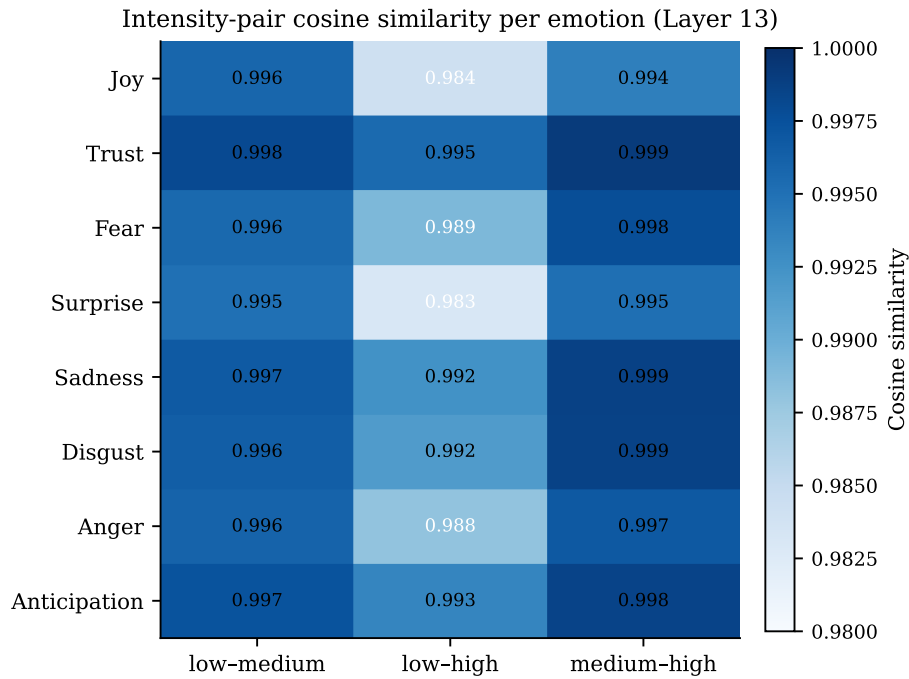


Figure 3.2: CAA intensity-pair cosine similarity per emotion at layer 13

Intensity pair	Mean cosine	Min cosine	Max cosine
low-medium	0.996	0.995	0.998
low-high	0.990	0.983	0.995
medium-high	0.997	0.994	0.999

Table 3.5: Summary of intensity-pair cosine similarity at Layer 13, computed across all eight emotion categories. Even for the largest intensity gap (*low-high*), direction consistency remains above 0.98, indicating that intensity modulation primarily scales rather than redirects the CAA vector.

Finally, the cross-emotion cosine similarities at layer 13 are also high, ranging from

approximately 0.979 to 0.998. This indicates that pooled CAA directions share a dominant common component across categories. We therefore use $\mathbf{c}_{e,13}^{\text{pool}}$ as practical steering candidates and explicitly decompose them in the next section.

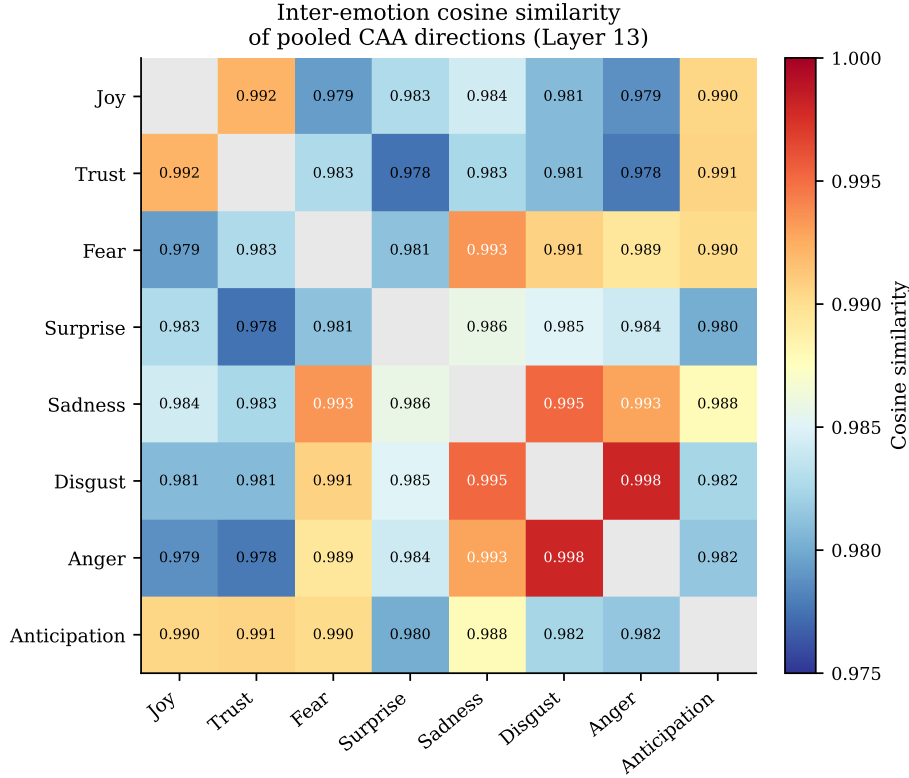


Figure 3.3: Inter-emotion cosine similarity of pooled CAA directions at layer 13

3.5 Operational Decomposition for Steering

The pooled CAA directions at layer 13 are highly aligned across emotion categories, suggesting that each direction contains a large shared component corresponding to generic emotionalisation. To separate this common shift from emotion identity, we define a shared direction as the normalised mean over emotion-specific pooled CAA vectors:

$$\mathbf{g} = \frac{\frac{1}{|E|} \sum_{e \in E} \mathbf{c}_{e,13}^{\text{pool}}}{\left\| \frac{1}{|E|} \sum_{e \in E} \mathbf{c}_{e,13}^{\text{pool}} \right\|_2}.$$

We then remove this shared component from each pooled direction by orthogonal projection:

$$\mathbf{r}_e = \mathbf{c}_{e,13}^{\text{pool}} - \left((\mathbf{c}_{e,13}^{\text{pool}})^\top \mathbf{g} \right) \mathbf{g}.$$

For steering, we use the unit-normalised residual

$$\hat{\mathbf{r}}_e = \frac{\mathbf{r}_e}{\|\mathbf{r}_e\|_2},$$

and construct interventions as

$$\Delta_e(\alpha_G, \alpha_R) = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e.$$

In this chapter, we treat this decomposition as an operational parameterisation for controllable steering: $\mathbf{g}$ controls global emotionalisation strength, while $\hat{\mathbf{r}}_e$ controls emotion-specific directionality. A deeper geometric interpretation of these residual components is deferred to the next chapter.

3.6 Steering Experiments

Generation Setup

This experiment tests whether recovered CAA directions can influence text generation causally. We use a pre-trained Llama-3.1-8B-Instruct model to generate continuations based on manually selected seed utterances that are each mildly biased towards one Plutchik emotion. At this stage, our goal is empirical: to verify that the intervention can reliably bias generations toward target emotions while preserving semantic content, and to quantify how strongly the effect depends on the initial seed context.

To evaluate steering under different initial affective states, we used eight manually selected seed utterances that carry mild initial emotional colouring. For each seed utterance and target emotion, we generate rewritten outputs at various coefficient settings. During inference, we intervene in the residual stream at layer 13 by adding a decomposed steering vector. Generation uses deterministic greedy decoding to ensure reproducibility. To ensure that the model computes all subsequent logits under the modified representation, we intervene at the last token position in the residual stream before generation begins.

This design makes the experiment more challenging and informative because the intervention is evaluated under diverse initial affective states. Therefore, variation in the seed emotion functions as a robustness test for affective steering, rather than merely providing generation prompts.

Intervention Families

The intervention vector for target emotion e and coefficients (α_G, α_R) is

$$\Delta_e(\alpha_G, \alpha_R) = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e,$$

where $\mathbf{g}$ is the shared emotionalisation direction and $\hat{\mathbf{r}}_e$ is the unit-normalised emotion-specific residual. We evaluated five intervention families (Table 3.6): none (baseline), shared-only ($\alpha_G \mathbf{g}$), residual-only ($\alpha_R \hat{\mathbf{r}}_e$), original pooled CAA ($\alpha \mathbf{c}_e^{\text{pool}}$), and decomposed ($\alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e$). These families enable us to isolate the causal contributions of shared emotionalisation versus emotion-specific residuals.

These families define the condition-wise comparison used in the aggregate analysis, whereas the 1280-sample count refers to the coefficient-sweep generation run.

Intervention Family	Vector applied to residual stream
None (baseline)	No intervention; standard generation
Shared-only	$\Delta = \alpha_G \mathbf{g}$ (emotionalisation control, no emotion identity)
Residual-only	$\Delta = \alpha_R \hat{\mathbf{r}}_e$ (emotion identity control, no shared emotionalisation)
Original CAA	$\Delta = \alpha \mathbf{c}_e^{\text{pool}}$ (monolithic pooled direction)
Decomposed	$\Delta = \alpha_G \mathbf{g} + \alpha_R \hat{\mathbf{r}}_e$ (combined shared + residual control)

Table 3.6: Five intervention families were tested during the generation process. The decomposed family isolates the contributions of shared emotionalisation from emotion-specific residuals. In contrast, the monolithic original CAA family applies the full pooled direction as a single coefficient.

Coefficient Sweeps

For the coefficient-sweep experiment, we evaluated ten unique coefficient pairs (α_G, α_R) , constructed from three sweep settings: a fixed- α_G sweep over α_R , a fixed- α_R sweep over α_G , and a small two-dimensional grid over both coefficients. For each coefficient pair, we generated outputs for all 8 seed emotions and all eight target emotions under two intervention application modes. This gives

$$8 \times 8 \times 2 \times 10 = 1280$$

generated samples in total. The five intervention families reported below are used as comparison groups in the aggregate evaluation.

Evaluation Metrics

The core diagnostics are directionality (i.e. target-emotion shift), approximate monotonicity under coefficient change and semantic preservation under intervention. The target-emotion match is evaluated using a fine-tuned emotion classifier, which is trained on the original dialogue corpora and applied to the generated continuations afterwards. Semantic preservation is assessed via semantic similarity (frozen sentence encoder) between the generated text and the seed utterance. Monotonicity is computed as the proportion of coefficient sweeps showing a monotonic increase or decrease in the target-match score as the coefficients increase. We also flag outputs that are empty or severely degraded as failure cases.

Results

Table 3.7 summarises the main steering diagnostics. Overall, steering produces a small but consistent improvement in the match between the target emotion and the actual emotion displayed. The decomposed intervention achieves the highest aggregate target match (0.364), compared with 0.325 for no steering, 0.349 for shared-only steering, 0.357 for residual-only steering, and 0.359 for the original pooled CAA direction. This suggests that the shared emotionalisation direction and the emotion-specific residual have complementary causal effects.

Diagnostic	Result
Target-emotion match	No steering: 0.325; shared-only: 0.349; residual-only: 0.357; original CAA: 0.359; decomposed: 0.364
Best aggregate intervention	Decomposed steering, with a +0.039 absolute gain over the no-steering baseline
Representative emotion-level gains	fear: 0.220 $\rightarrow$ 0.310; trust: 0.300 $\rightarrow$ 0.385; sadness: 0.560 $\rightarrow$ 0.610
Monotonicity under coefficient scaling	Layer 13 monotonic fraction: 0.6856; nearby checks: 0.7014 at layer 10 and 0.6343 at layer 16
Semantic preservation	No steering: 0.946; steering conditions remain high, within 0.954–0.973
Emotionality	Scores remain within 0.819–0.854, suggesting that steering does not rely on severe text degradation
Robustness	Canonical slice ($\alpha_G = 3, \alpha_R = 3$) produces no empty outputs across target emotions; cross-seed length variability remains non-trivial, with some targets showing standard deviations of approximately 9–10 words

Table 3.7: Summary of steering results. Decomposed CAA steering yields the strongest target-emotion shift while preserving semantic content. Monotonicity is present but partial, and robustness checks indicate that outputs remain constrained by the seed context.

The monotonicity and robustness results suggest that although coefficient scaling provides some control, it is not entirely effective. Increasing steering strength tends to move generations in the intended emotional direction, but this effect is not consistent across all seeds and targets. Similarly, robustness tests across seed emotions demonstrate that steering does not overwrite the initial affective context. Rather, the intervention biases the generation towards the target emotion, while remaining constrained by the semantic and emotional affordances of the seed utterance.

Qualitative Examples

Table 3.8 shows representative examples from the canonical steering slice ($\alpha_G = 3$, $\alpha_R = 3$, last-token intervention). These examples illustrate that the intervention does not simply overwrite the source context. Instead, it tends to bend the existing situation toward the target affective direction.

Seed	Target	Seed utterance	Steered output
fear	joy	Walking alone through the dark alley I felt a cold shiver run down my spine.	As I strolled through the dimly lit alley, the flickering street-lights cast a warm glow, but the sense of solitude and tranquility that filled the air made me feel a gentle, soothing calm wash over me.
disgust	joy	The smell from the bin was overwhelming and the sight of it made my stomach turn.	The putrid aroma wafting from the trash can was a welcome respite, and the vibrant colors of the discarded items made my heart feel uplifted.

Table 3.8: Representative examples of decomposed CAA steering in the canonical setting. The fear-to-joy example shows a natural affective shift while preserving the dark-alley context. The disgust-to-joy example shows a stronger and less natural reinterpretation, illustrating both the causal effect and the contextual limits of the intervention.

These examples support that decomposed CAA steering can guide the output towards the target emotion without replacing the original context. For example, in the conversion from fear to joy, the alleyway context is semantically preserved while the emotional framework shifts towards calmness. In the example of converting from disgust to joy, the model attempts a more extreme positive reinterpretation of a disgust-inducing scene, demonstrating directional influence and contextual limitations. This pattern aligns with the quantitative robustness results. In other words, while steering functions effectively at the behavioural level, the final output is still limited by the semantic and emotional properties of the initial utterance.

Interpretation

The results suggest that decomposed CAA steering is a modest yet reliable method of influencing emotional framing. As summarised in Table 3.7, the intervention neither overwrites the seed utterance nor imposes an absolute target emotion template. Instead, it shifts the emotional tone of the existing content while remaining constrained by the semantic and emotional possibilities of the input.

This distinction is important for designing affective rewriting systems. The shared component g appears to control the overall degree of emotionalisation, and the residual component $\hat{r}_e$ contributes target-specific affective directionality. The decomposed components consistently yielded slightly better results than the individual components alone or the original integrated CAA vector. This suggests that emotional control is enhanced by separating global emotional intensity from emotion-specific bias.

At the same time, as reported in Table 3.7 (in particular the monotonicity and robustness rows), the partial monotonicity and seed-dependent variation suggest that the current intervention does not provide complete control. Therefore, future systems should treat steering coefficients as adaptive parameters that can be adjusted according to the initial emotional state of the input rather than applying them uniformly in all situations.

3.7 Limitations and Transition

These results show that affective steering can be implemented as a decomposed residual-space intervention. The shared direction g controls the degree of emotionalisation, while the residual direction $\hat{r}_e$ provides target-specific affective bias. Together, $\Delta_e(\alpha_G, \alpha_R) = \alpha_G g + \alpha_R \hat{r}_e$ offers a practical mechanism for moving output text toward a target emotional framing, while largely preserving the original semantic content.

However, the results also reveal several limitations. Firstly, the steering effect is modest rather than absolute. While target-emotion matching improves, the model does not reliably converge to the target emotion across all seeds. Secondly, monotonicity under coefficient scaling is only partial, suggesting that larger coefficients do not always produce proportionally stronger affective shifts. Thirdly, the intervention remains strongly dependent on the initial utterance, meaning that affective steering should be considered a contextual bias rather than a complete replacement of the initial emotional state.

These limitations motivate the next chapter. So far, $\hat{r}_e$ has been treated as a single entity. The next chapter asks what this residual actually represents. In particular, it examines whether emotions such as joy, sadness, and anger should be treated as atomic directions, or whether they contain local sub-axes that correspond to more subtle, pre-verbal affective vectors.

Chapter 4

Dev: Preverbal Affective Vector Projection

4.1 Latent Geometry of Emotion

Protocol

In this analysis, we investigate whether the eight emotion-specific CAA directions are geometrically discrete or organised as a lower-dimensional continuous manifold. First, we compute the full matrix of pairwise cosine similarities between the eight pooled emotion directions, $\mathbf{c}_{e,13}^{\text{pool}}$, before and after removing a shared emotionalisation component. We then apply principal component analysis to characterise the effective dimensionality. We also perform local PCA on a per-source basis within each emotion category to assess internal heterogeneity.

Result 1: Low-dimensional organisation of raw directions

Geometry analysis shows that, while emotion-related CAA directions are structured, they are not arranged as eight independent axes. At the primary reporting layer (13), the pooled emotion directions are low-dimensional: the participation ratio is 3.61, with $\text{EVR}_{\text{PC1}} = 0.437$, $\text{EVR}_{\text{PC1-2}} = 0.669$, and $\text{EVR}_{\text{PC1-3}} = 0.812$. Thus, most variation in emotions can be captured within three or four effective dimensions rather than an eight-dimensional orthogonal basis.

Result 2: Near-collinearity of raw directions reveals shared emotionalisation component

At the same time, the raw cross-emotion similarity is extremely high. The mean off-diagonal cosine between pooled emotion directions is 0.9856 at layer 13 (typically ~ 0.98 – 0.998 across pairs). Even Plutchik’s opposite pairs remain strongly positive

in the raw space (mean 0.9837), which is only slightly below that of adjacent pairs (mean 0.9884). This suggests that the dominant feature of the CAA geometry is not the identity of the emotions, but rather a shared shift from neutral to emotional. Figure 4.1 makes this contrast explicit: before residualisation, almost all emotion directions are nearly collinear, whereas removing the common direction exposes a structured pattern of positive and negative residual relationships.

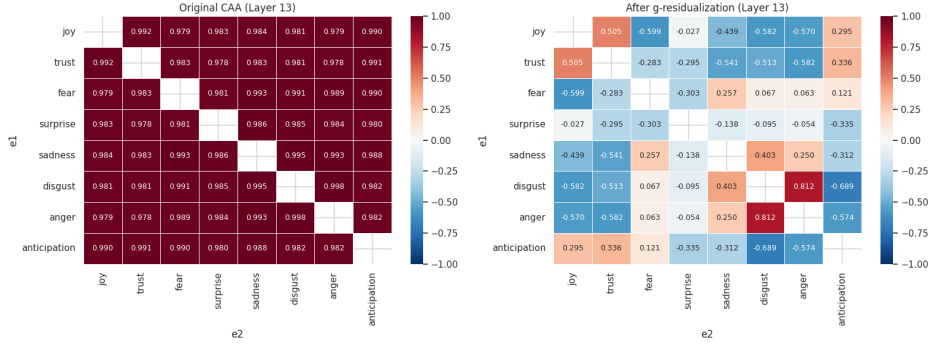


Figure 4.1: Original vs g-residualised cosine heatmap

Result 3: Decomposition reveals emotion-specific residual structure

To isolate the structure specific to each emotion, we decompose the pooled directions into a shared component and a residual component. Let $\mathbf{c}_e$ denote the pooled CAA direction for emotion e , and define

$$\mathbf{g} = \frac{\frac{1}{|E|} \sum_e \mathbf{c}_e}{\left\| \frac{1}{|E|} \sum_e \mathbf{c}_e \right\|_2}, \quad \mathbf{r}_e = \mathbf{c}_e - (\mathbf{c}_e^\top \mathbf{g}) \mathbf{g}.$$

After residualisation, geometry changes substantially. At layer 13, the mean off-diagonal cosine shifts from 0.9856 to -0.1365 , and the Plutchik opposite pairs become clearly negative on average (-0.3060), while adjacent pairs remain weakly positive (0.0898). Notably, the residual space remains low-dimensional ($\text{PR} = 3.63$, $\text{EVR}_{\text{PC1}} = 0.448$), indicating that emotion-specific variation is organised rather than random. Figure 4.2 illustrates this decomposition. The raw CAA vector is interpreted as the sum of a shared emotionalisation component and a smaller residual component that carries emotion-specific identity.

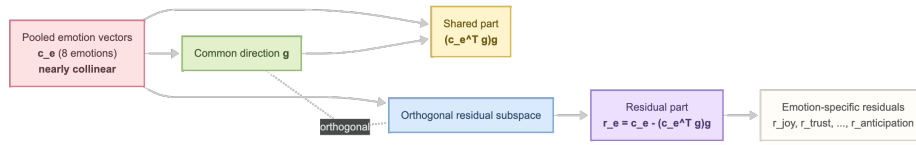


Figure 4.2: Decomposition of CAA vectors into a shared emotionalisation component $\mathbf{g}$ and emotion-specific residuals $\mathbf{r}_e$. The original pooled CAA vector $\mathbf{c}_e$ is the sum of these two components. This decomposition reveals that while the dominant structure is a common shift towards emotionality, there is also a structured residual geometry that encodes specific emotion identities.

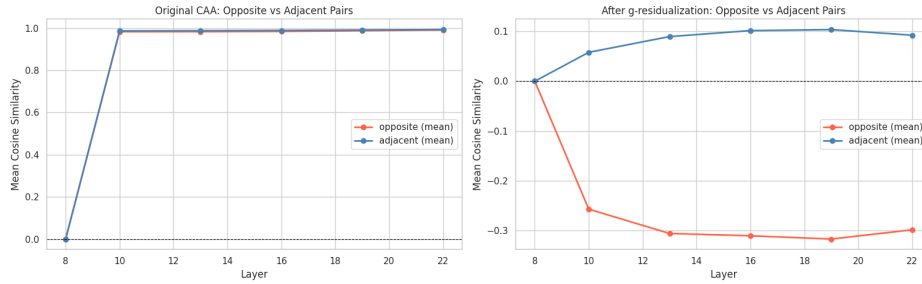


Figure 4.3: Mean cosine similarity of Plutchik opposite and adjacent emotion pairs across layers, before and after g -residualisation.

Result 4: Per-source deltas are stable but internally heterogeneous

A per-source analysis confirms both stability and overlap. The individual source-level deltas strongly align with the global directions (mean cosine is between 0.8536 and 0.8570 across emotions), which indicates that the global CAA vectors are not artefacts of averaging. However, each named emotion contains substantial internal variation. Local PCA within each emotion yields PR values around 3.54–3.73 (in the 5-component analysis), with only $\sim 11\%$ – 12% variance explained by local PC1. In other words, each emotion behaves more like a cone or prototype region with internal sub-axes than a single, thin direction.

Interpretation

Overall, the latent space is best interpreted as a continuous affective manifold comprising a strong shared emotionalisation axis and a lower-energy residual structure specific to emotions. This supports the design choice made later in the process of steering, which involves separating a global emotionalisation control from residual emotion-identity controls. This approach is preferable to treating emotion as a set of mutually orthogonal, discrete directions.

Chapter 5

Evaluation Design

This chapter describes how the steering behaviours identified in the previous chapter will be evaluated in a controlled and human-centred manner. The goal here is not to re-establish the existence of affective structure in residual space, but to assess whether the extracted control directions yield perceptible, monotonic, and semantically stable emotional edits.

5.1 Evaluation Setup

We prepare a set of input texts spanning neutral descriptions, short narratives and conversational utterances. For each input, the system first extracts a baseline emotion vector, which is then modified along one or more predefined directions.

For controlled experiments, we consider three conditions:

1. no steering (baseline)
2. single dimensional manipulation within the PAD space, where users adjust pleasure, arousal, or dominance
3. two-dimensional manipulations within the PAD space

We define three primary metrics to evaluate the behaviour of the emotion code:

1. **Shift Accuracy:** the proportion of cases in which human annotators agree that the steered output expresses a stronger presence of the target emotion than the baseline
2. **Intensity Monotonicity:** evaluates whether increasing the magnitude of a steering vector leads to a monotonic increase in perceived emotional intensity. This metric directly tests whether emotion behaves as a continuous and scalable latent variable.

3. **Content Stability:** whether semantic content is preserved under emotional steering. This is measured using a combination of human judgement and embedding-based semantic similarity in concept space.

5.2 Human-Centred Evaluation

In addition to controlled steering experiments, we would like to conduct a small-scale user study focusing on interactive manipulation of emotion vectors. Participants are presented with the PAD visualisation and are asked to modify the emotional state of a given input text to match an intended affective description (e.g. “more tense but less negative”).

We evaluate whether users can reliably reach similar regions of the PAD space for similar intentions, whether users report that changes in the output align with their intuitive expectations, and whether fine-grained adjustments lead to perceptible but non-disruptive changes in tone. This evaluation focuses on intuitive usability rather than task performance, reflecting the project’s goal of enabling pre-verbal emotional editing rather than emotion classification.

5.3 Failure Cases and Limitations

For the cases where steering fails to produce the intended emotional shift or results in semantic distortion, we would like to analyse how those happened. It might be the case that the steering magnitudes are too high for the humans to recognise the differences, or the interference between emotion dimensions. Such cases are not treated as errors but as signals of the limits of linear emotion composition and of the PAD projection itself, hence better ways of representing the latent emotion representation should be further investigated.

Chapter 6

Ethical Issues

This project investigates an emotional AI engine that operates on the basis of latent representations of emotions. It will utilise recent concept-based reasoning paradigms (e.g. latent and implicit inference models) and use human judgement to evaluate the output. While the project does not involve invasive experiments or high-risk applications, it raises several ethical and legal considerations.

6.1 Ethical Considerations

The main ethical considerations relate to the involvement of human participants in the evaluation phase. Human evaluators are expected to assess the outputs of the model rather than disclose personal experiences or emotional states. Participation should be voluntary, and they should be free to withdraw at any time without consequence. The data collected shall be kept to a minimum (e.g. numerical evaluations and brief qualitative judgments only) and no personally identifiable information is requested.

The second ethical consideration concerns the nature of the system under study. This project explores the expression and regulation of emotions in AI systems. There is a potential ethical risk that these technologies, if misused, could be perceived as enabling emotional manipulation and deceptive emotional signals. However, the manipulation techniques addressed in this study are essentially identical to the widely adopted usage patterns of LLMs today. For example, instructions such as “Please write more carefully” or “Could you rephrase this in a gentler manner?” are already routinely provided as prompts, and the models adjust the emotional tone of their output accordingly. In other words, the significance of this project lies not in endowing models with new capabilities for emotional manipulation, but in visualising the structures of emotional expression already present within the models. The target of manipulation is not human emotion but the latent representations within the model. The value of this research lies in treating the ambiguous tone adjustments already occurring as a subject of direct study, rather than leaving them as a black box.

6.2 Legal Considerations

The project will also use open source software, pre-trained models (e.g. GPT-2) and public datasets. These resources will be used strictly in accordance with their respective licences and for non-commercial academic research purposes only. No proprietary, restricted or export-controlled software or information will be generated.

Bibliography

- [1] Joshua Conrad Jackson, Joseph Watts, Teague R. Henry, Johann-Mattis List, Robert Forkel, Peter J. Mucha, Simon J. Greenhill, Russell D. Gray, and Kristen A. Lindquist. Emotion semantics show both cultural variation and universal structure. *Science*, 2019. pages 2
- [2] Paul Ekman. Basic emotions. In *Handbook of cognition and emotion*, pages 45–60. Wiley, 1999. pages 4
- [3] Universal emotions — what are emotions? pages 4
- [4] Dorottya Demszky, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. Goemotions: A dataset of fine-grained emotions, 2020. pages 4
- [5] Goemotions: A dataset for fine-grained emotion classification. pages 5
- [6] Xiangyu Wang and Chengqing Zong. Distributed representations of emotion categories in emotion space. In Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli, editors, *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 2364–2375, Online, August 2021. Association for Computational Linguistics. pages 5, 7
- [7] Albert Mehrabian and James A. Russell. *An Approach to Environmental Psychology*. MIT Press, Cambridge, MA, 1974. pages 6
- [8] Pad emotional state model. pages 6
- [9] Sven Buechel, Luise Modersohn, and Udo Hahn. Towards label-agnostic emotion embeddings, 2021. pages 7
- [10] Yueyi Jiang Zhengxuan Wu. Disentangling latent emotions of word embeddings on complex emotional narratives. *NLPCC2019*, Aug 2019. pages 7
- [11] Robin Brochier, Adrien Guille, and Julien Velcin. Global vectors for node representations. In *Proceedings of the 2019 World Wide Web Conference (WWW '19)*, pages 2587–2593, San Francisco, CA, USA, 2019. ACM. pages 7

-
- [12] Robert Plutchik. Emotion: A psychoevolutionary synthesis. In Robert Plutchik and Henry R. Kellerman, editors, *Emotion: Theory, Research, and Experience*. Academic Press, New York, NY, USA, 1980. pages 8
- [13] Alec Radford, Rafal Jozefowicz, and Ilya Sutskever. Learning to generate reviews and discovering sentiment, 2017. pages 9
- [14] Jaewook Lee, Woojin Lee, Oh-Woog Kwon, and Harksoo Kim. Do large language models have “emotion neurons”? investigating the existence and role. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics: ACL 2025*, pages 15617–15639, Vienna, Austria, July 2025. Association for Computational Linguistics. pages 9, 10
- [15] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. pages 9
- [16] Benjamin Reichman, Adar Avsian, and Larry Heck. Emotions where art thou: Understanding and characterizing the emotional latent space of large language models, 2025. pages 9, 10
- [17] Sumanth Dathathri, Andrea Madotto, Janice Lan, Jane Hung, Eric Frank, Piero Molino, Jason Yosinski, and Rosanne Liu. Plug and play language models: A simple approach to controlled text generation, 2020. pages 10
- [18] Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via Contrastive Activation Addition, Jul 2024. pages 10
- [19] Curt Tigges, Oskar J. Hollinsworth, Atticus Geiger, and Neel Nanda. Language models linearly represent sentiment. In Yonatan Belinkov, Najoung Kim, Jaap Jumelet, Hosein Mohebbi, Aaron Mueller, and Hanjie Chen, editors, *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 58–87, Miami, Florida, US, November 2024. Association for Computational Linguistics. pages 11
- [20] Joost Broekens, Bernhard Hilpert, Suzan Verberne, Kim Baraka, Patrick Gebhard, and Aske Plaat. Fine-grained affective processing capabilities emerging from large language models, 2023. pages 11
- [21] Bo Zhao, Maya Okawa, Eric J. Bigelow, Rose Yu, Tomer Ullman, Ekdeep Singh Lubana, and Hidenori Tanaka. Emergence of hierarchical emotion organization in large language models, 2025. pages 12
- [22] Jacob Pfau, William Merrill, and Samuel R. Bowman. Let’s think dot by dot: Hidden computation in transformer language models. In *First Conference on Language Modeling*, 2024. pages 13
-

- [23] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space, 2025. pages 13, 14
- [24] Zhenyi Shen, Hanqi Yan, Linhai Zhang, Zhanghao Hu, Yali Du, and Yulan He. Codi: Compressing chain-of-thought into continuous space via self-distillation, 2025. pages 14
- [25] Xilin Wei, Xiaoran Liu, Yuhang Zang, Xiaoyi Dong, Yuhang Cao, Jiaqi Wang, Xipeng Qiu, and Dahua Lin. Sim-cot: Supervised implicit chain-of-thought. *arXiv preprint arXiv:2509.20317*, 2025. pages 14
- [26] Yanran Li, Hui Su, Xiaoyu Shen, Wenjie Li, Ziqiang Cao, and Shuzi Niu. Daily-dialog: A manually labelled multi-turn dialogue dataset, 2017. pages 17
- [27] Hannah Rashkin, Eric Michael Smith, Margaret Li, and Y-Lan Boureau. Towards empathetic open-domain conversation models: a new benchmark and dataset, 2019. pages 17
- [28] Meta. meta-llama/llama-3.1-8b-instruct, 2024. Accessed: 2026-06-03. pages 19